

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Project Description:

This project streamlines IT procurement by automating the ordering of standard laptops on the ServiceNow platform. Employees request a laptop through a Service Catalog item, and a Flow Designer flow takes over: it routes the request to the employee's manager for approval, creates a fulfillment task for the IT team after approval, updates the request state, and sends email notifications at each stage. The platform includes a Standard Laptop catalog item, an automated approval workflow, and a status report that tracks turnaround time.

Built with ServiceNow Service Catalog and Flow Designer, the solution replaces manual email chains and spreadsheets with a low-code workflow. Role-based access keeps requesters, approvers and fulfillers separate, and every request is recorded in platform tables, giving the IT team a reliable audit trail and giving employees clear visibility of their order.

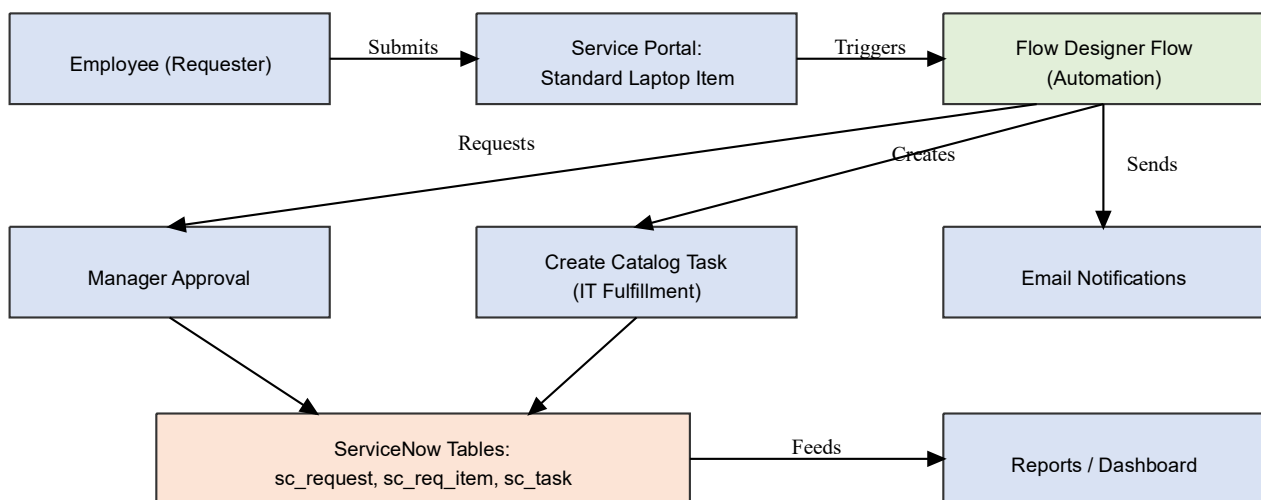
Scenarios

Scenario 1: A new employee joins the company and needs a laptop. They open the Service Portal, select the Standard Laptop item, choose the model and delivery location, and submit. The manager receives an approval automatically, and once approved, IT gets a fulfillment task without anyone sending an email.

Scenario 2: An employee requests a laptop that the manager decides to reject. The manager rejects the approval, the flow closes the request with a rejected state, and the employee is notified immediately with no manual follow-up from IT.

Scenario 3: An IT manager wants to know how quickly laptop requests are being fulfilled. They open the request status report and dashboard to see open, approved and completed requests and the average turnaround time.

Technical Architecture:



Description: The solution follows a layered architecture on the ServiceNow platform. The presentation layer is the Service Portal and Service Catalog, where employees submit requests. The automation layer is a Flow Designer flow that handles approval, task creation, state updates and notifications. The data layer consists of ServiceNow tables (sc_request, sc_req_item, sc_task) that store every request, and the reporting layer reads from these tables to show status and turnaround time.

(Note: ServiceNow provides its own relational database, so no separate database or ER diagram is required for this project. The key tables are listed below.)

Table	Purpose
sc_request	Parent request created when the laptop is ordered
sc_req_item	Requested item (Standard Laptop) with its variables and state
sc_task	Catalog task assigned to the IT fulfillment group
sysapproval_approver	Approval record for the manager
sys_user / sys_user_group	Users, managers and IT groups

Pre-requisites:

- **ServiceNow Platform Knowledge:** [ServiceNow Documentation](#)
- **ServiceNow Developer Program (free Personal Developer Instance):** [ServiceNow Developer Site](#)
- **Service Catalog Basics:** catalog items, variables and requested items
- **Flow Designer Basics:** triggers, actions, approvals and flow logic
- **Users, Groups and Roles:** creating test users and assigning roles
- **Update Sets:** exporting and importing configuration between instances
- **Web Browser:** Google Chrome or Microsoft Edge

Project Workflow:

Milestone 1: Platform Setup and Architecture

- Activity 1.1: Create a ServiceNow Personal Developer Instance and log in as admin.
- Activity 1.2: Identify the standard laptop models and the approval rules for the request process.
- Activity 1.3: Define the architecture of the solution, detailing the catalog, flow, tables and notifications.
- Activity 1.4: Set up users, groups and roles for requesters, managers and IT fulfillers.

Milestone 2: Service Catalog Development

- Activity 2.1: Create the Standard Laptop catalog item and its variables.
- Activity 2.2: Publish the item to the Service Catalog and make it available in the Service Portal.

Milestone 3: Flow Designer Development

- Activity 3.1: Create the flow with a Service Catalog trigger.
- Activity 3.2: Add manager approval, catalog task creation and request state updates.

Milestone 4: Notifications and Reporting

- Activity 4.1: Configure email notifications for submission, approval, rejection and fulfillment.
- Activity 4.2: Build a request status report and dashboard.

Milestone 5: Testing and Deployment

- Activity 5.1: Test the approved and rejected paths end to end.
- Activity 5.2: Export the Update Set for deployment to another instance.

Milestone 6: Conclusion

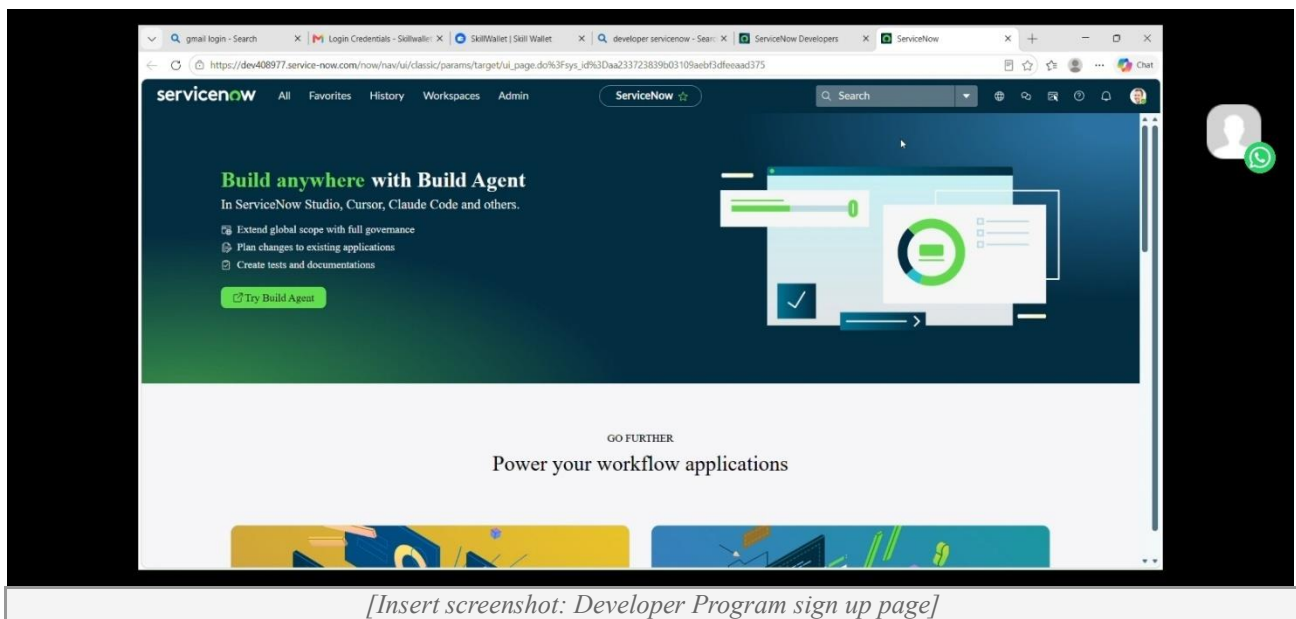
Milestone 1: Platform Setup and Architecture

In this milestone, we set up the ServiceNow environment and define how the catalog, flow and data layers work together to automate standard laptop orders.

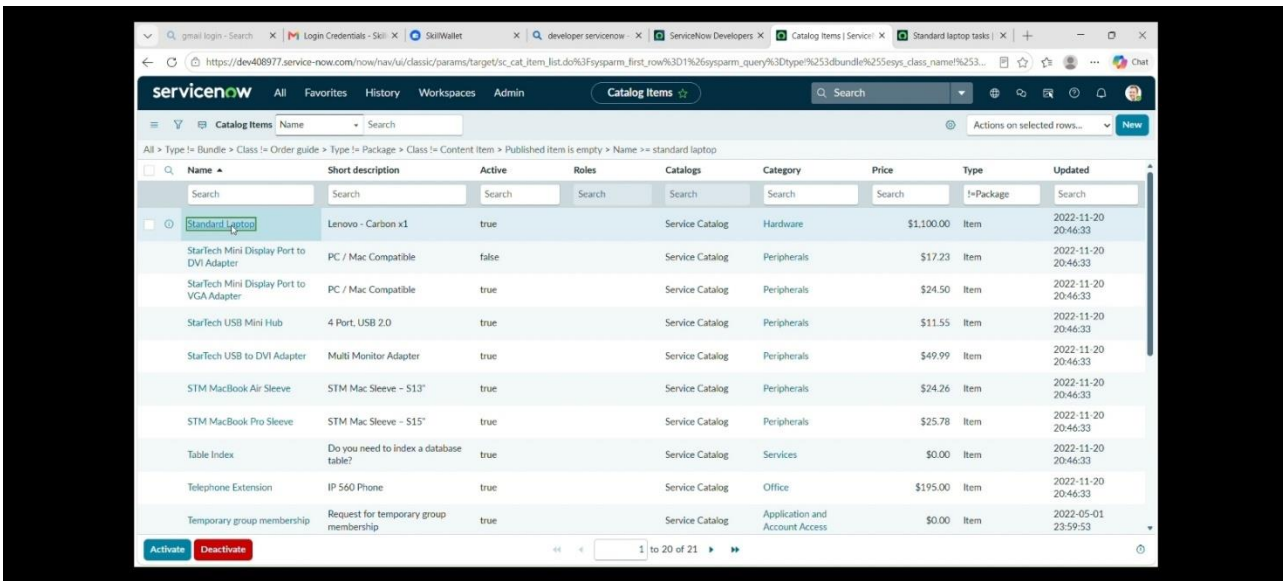
Activity 1.1: Create a ServiceNow Personal Developer Instance

A Personal Developer Instance (PDI) is a free ServiceNow instance for learning and building. Follow the steps below to get one.

- **Step 1 — Create a Developer Account:** Visit developer.servicenow.com and sign up with your email address.



- **Step 2 — Request an Instance:** After logging in, open your profile menu, choose Request Instance and select the latest release.
- **Step 3 — Open the Instance:** When the instance is ready, note the instance URL, the admin username and the password shown on screen, and click Open Instance.
- **Important — Keep Credentials Safe:** Do not share the admin password. Create separate demo users for testing and for the evaluator.



The screenshot shows the ServiceNow Catalog Items list. The table contains the following data:

Name	Short description	Active	Roles	Catalogs	Category	Price	Type	Updated
Standard Laptop	Lenovo - Carbon x1	true		Service Catalog	Hardware	\$1,100.00	Item	2022-11-20 20:46:33
StarTech Mini Display Port to DVI Adapter	PC / Mac Compatible	false		Service Catalog	Peripherals	\$17.23	Item	2022-11-20 20:46:33
StarTech Mini Display Port to VGA Adapter	PC / Mac Compatible	true		Service Catalog	Peripherals	\$24.50	Item	2022-11-20 20:46:33
StarTech USB Mini Hub	4 Port USB 2.0	true		Service Catalog	Peripherals	\$11.55	Item	2022-11-20 20:46:33
StarTech USB to DVI Adapter	Multi Monitor Adapter	true		Service Catalog	Peripherals	\$49.99	Item	2022-11-20 20:46:33
STM MacBook Air Sleeve	STM Mac Sleeve - S13"	true		Service Catalog	Peripherals	\$24.26	Item	2022-11-20 20:46:33
STM MacBook Pro Sleeve	STM Mac Sleeve - S15"	true		Service Catalog	Peripherals	\$25.78	Item	2022-11-20 20:46:33
Table Index	Do you need to index a database table?	true		Service Catalog	Services	\$0.00	Item	2022-11-20 20:46:33
Telephone Extension	IP 560 Phone	true		Service Catalog	Office	\$195.00	Item	2022-11-20 20:46:33
Temporary group membership	Request for temporary group membership	true		Service Catalog	Application and Account Access	\$0.00	Item	2022-05-01 23:59:53

Activity 1.2: Identify Requirements

- **Understand the Process:** Review how laptop orders are handled today (email requests, manual approvals, spreadsheet tracking).
- **Define Standard Models:** Decide which laptop models employees may order (for example Standard Business Laptop and Developer Laptop).
- **Define the Approval Rule:** The requester's manager approves every standard laptop request.
- **Define Fulfillment:** An IT Procurement group receives a task after approval and delivers the laptop.

Activity 1.3: Define the Architecture of the Application

- **Draft an Architectural Diagram:** Show the Service Portal, catalog item, Flow Designer flow, tables, notifications and reports.
- **Detail Requester Functionality:** The employee opens the catalog, selects the laptop model, enters justification and delivery location, and submits.
- **Outline Flow Responsibilities:** The flow gets the catalog variables, asks the manager for approval, creates a catalog task on approval, updates the request state and sends emails.
- **Describe Data Storage:** All requests, items, tasks and approvals are stored in ServiceNow tables and used for reporting.

Activity 1.4: Set Up Users, Groups and Roles

- **Create Users:** Go to All > User Administration > Users and create a demo requester, a demo manager and a demo IT fulfiller. Set the manager field on the requester's user record.
- **Create the Fulfillment Group:** Go to User Administration > Groups and create a group named IT Procurement. Add the IT fulfiller as a member.

- **Assign Roles:** Give the requester the base user role, the manager the approver role needs, and the fulfiller the catalog task fulfiller role, following your organization's role naming.

User	Role in the Process	Group
Demo Requester	Orders the laptop	—
Demo Manager	Approves or rejects the request	—
Demo Fulfiller	Completes the catalog task	IT Procurement

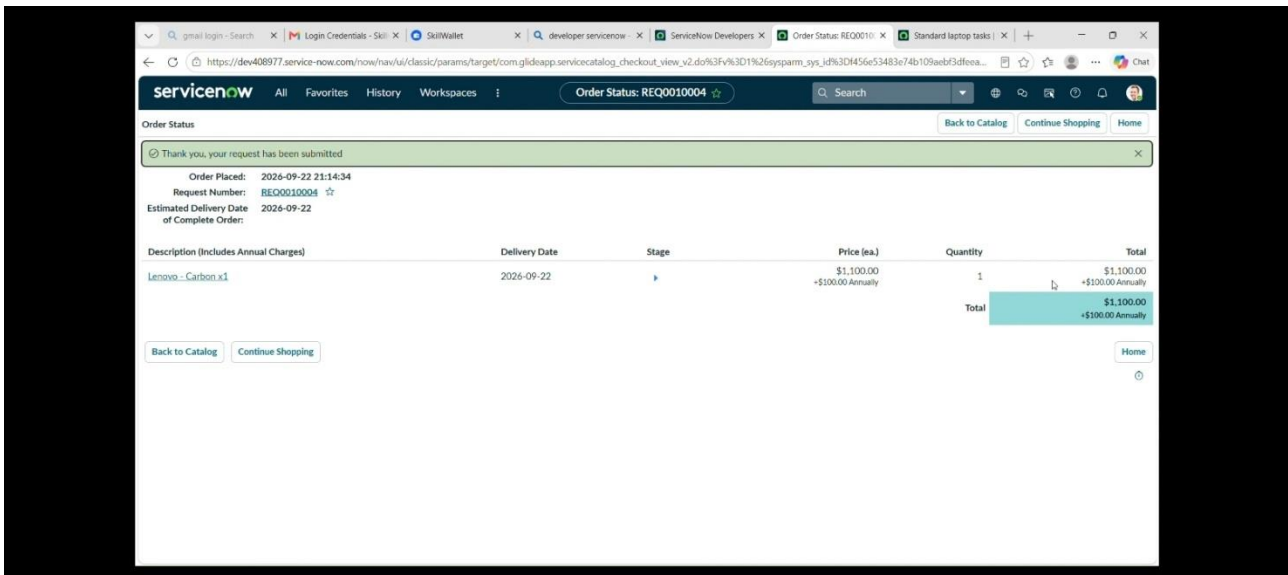
[Insert screenshot: Users list and IT Procurement group members]

Milestone 2: Service Catalog Development

This milestone creates the Standard Laptop catalog item that employees use to request a laptop. The item collects only the information IT needs, so requests arrive complete and consistent.

Activity 2.1: Create the Standard Laptop Catalog Item

- **Step 1 — Open Catalog Items:** Go to All > Service Catalog > Catalog Definitions > Maintain Items and click New.
- **Step 2 — Enter Item Details:** Set Name to Standard Laptop, choose a Catalog and Category (for example Hardware), and add a short description and price if required.
- **Step 3 — Save the Item:** Click Submit so that the Variables related list becomes available.



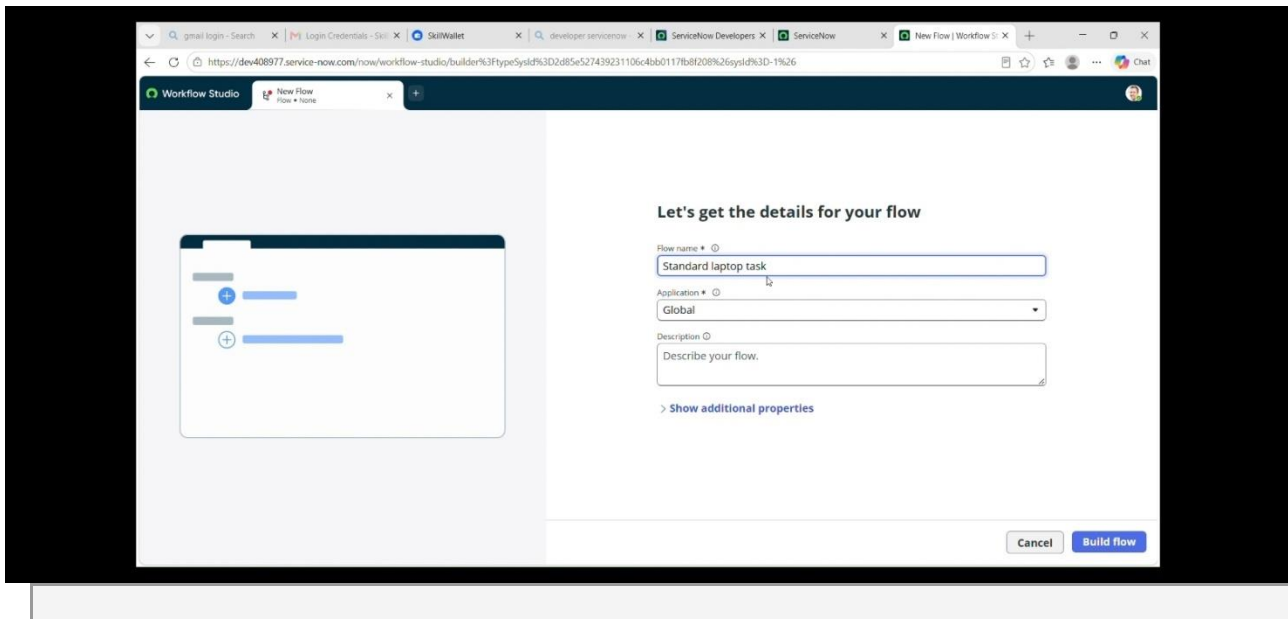
Activity 2.2: Add Variables and Publish

Open the catalog item and add the variables below in the Variables related list.

Variable Name	Type	Purpose
laptop_model	Select Box	Choose an approved standard model
business_justification	Multi Line Text	Reason for the request

delivery_location	Single Line Text	Where the laptop should be delivered
needed_by	Date	Date the laptop is needed

- **Make Required Fields Mandatory:** Mark laptop_model and delivery_location as mandatory so incomplete requests cannot be submitted.
- **Publish to the Portal:** Make sure the item is Active and the category is available in the Service Portal (/sp) so employees can find it.



Milestone 3: Flow Designer Development

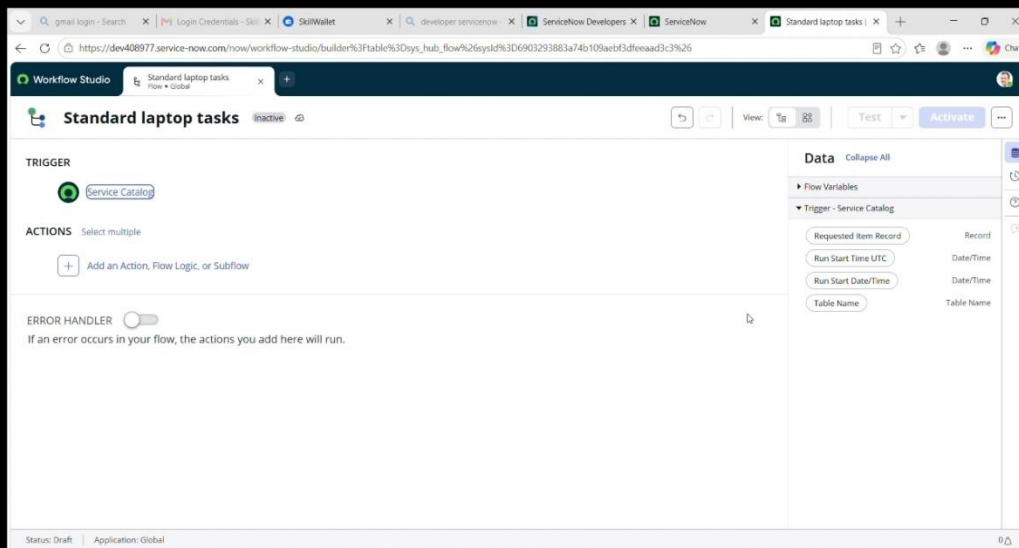
Flow Designer is ServiceNow's low-code automation tool. In this milestone we build the flow that runs every time a Standard Laptop is requested, replacing manual follow-ups.

Activity 3.1: Create the Flow and Trigger

- **Step 1 — Open Flow Designer:** Go to All > Process Automation > Flow Designer and click New > Flow.
- **Step 2 — Name the Flow:** Enter Standard Laptop Order Flow with a short description and click Build flow.
- **Step 3 — Add the Trigger:** Choose the Service Catalog trigger and select the Standard Laptop catalog item so the flow starts for every request of this item.

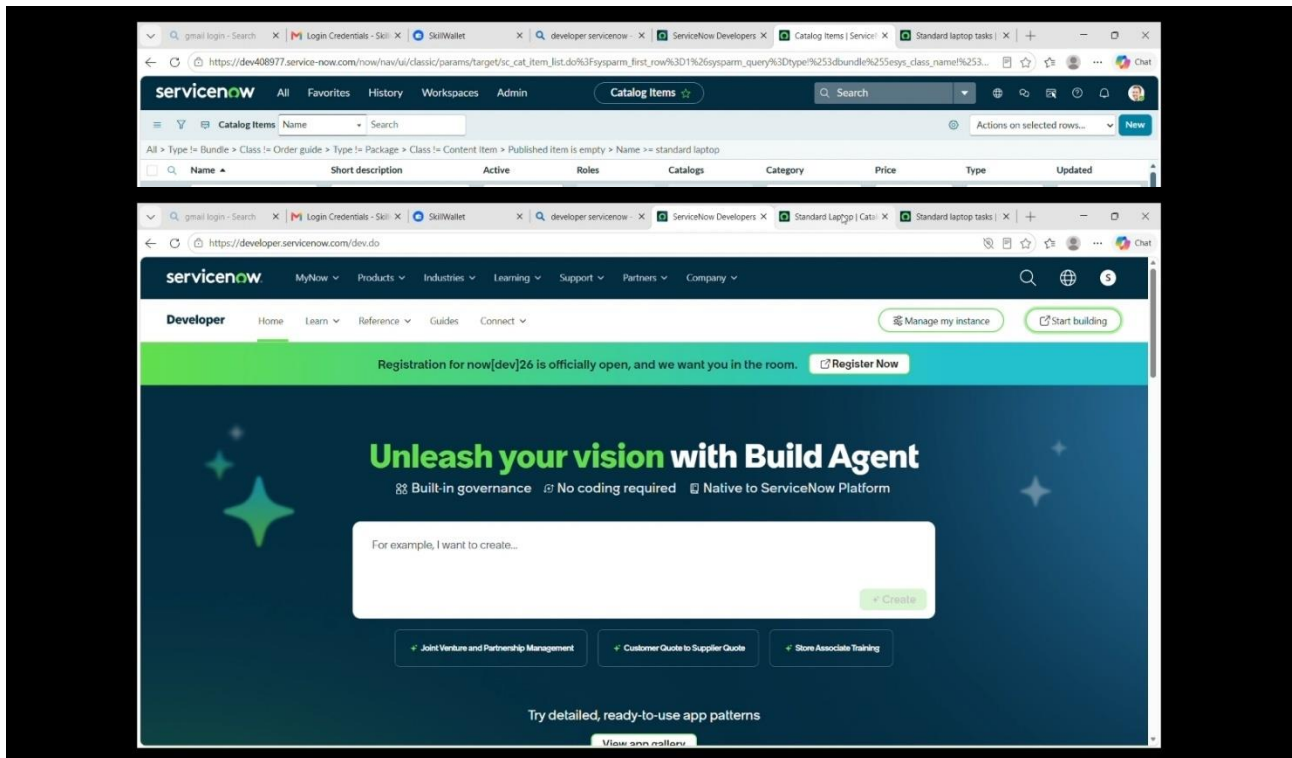
Activity 3.2: Add Approval, Task Creation and State Updates

Add the actions below in order. Data pills from the trigger (such as the requested item and the requester's manager) are dragged into the action inputs.



Order	Flow Step	What It Does
1	Get Catalog Variables	Reads the model, justification and delivery location
2	Ask For Approval	Sends the request to the requester's manager
3	If approved	Continues with fulfillment
4	Create Catalog Task	Creates a task for the IT Procurement group with the model and delivery location
5	Update Record	Sets the requested item state to Work in Progress
6	Send Email	Tells the requester the order was approved
7	Else (rejected)	Updates the record to Closed Incomplete and emails the requester

- **Ask For Approval:** Set the record to the requested item, the approver to the requester's manager, and the rule to approve when the manager approves.
- **Create Catalog Task:** Assign the task to the IT Procurement group and include the laptop model and delivery location in the description.
- **Activate the Flow:** Click Save, then Activate so it runs on new requests.



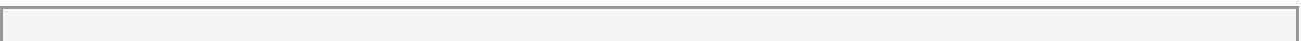
• Milestone 4: Notifications and Reporting

Notifications keep employees informed, and reports give the IT team visibility of open requests and turnaround time.

Activity 4.1: Configure Email Notifications

- **Submission Confirmation:** Notify the requester when the request is created.
- **Approval Request:** Notify the manager that an approval is waiting.
- **Approved / Rejected:** Notify the requester of the manager's decision, using the Send Email actions in the flow.
- **Fulfilled:** Notify the requester when the catalog task is closed complete.

Emails can be reviewed under All > System Logs > Emails. On a developer instance, delivery of outbound email may be limited, so use this log to verify that notifications were generated.



Activity 4.2: Build the Request Status Report and Dashboard

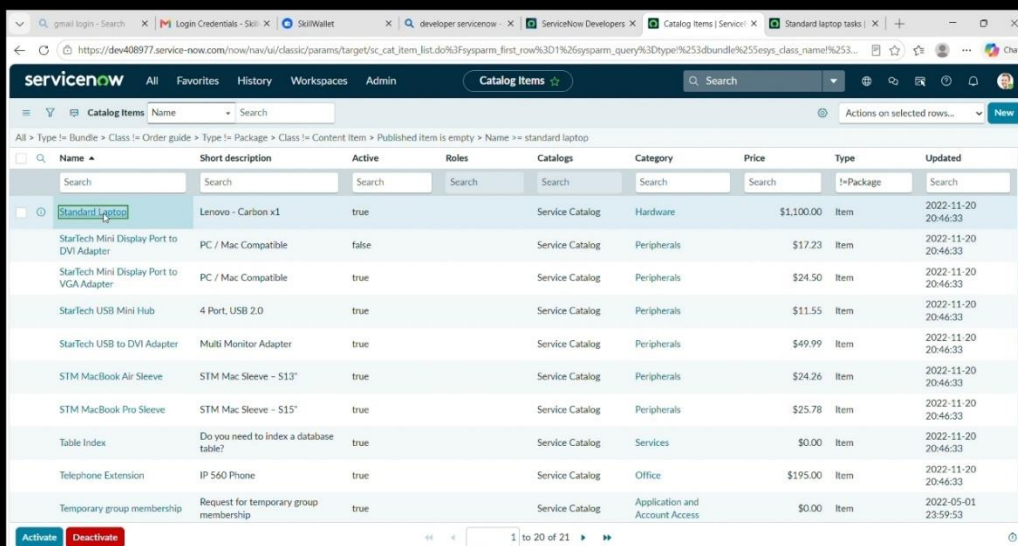
- **Create a Report:** Go to All > Reports > Create New, choose the Requested Item table (sc_req_item), filter on the Standard Laptop item and group by State.
- **Choose a Chart:** Use a bar or pie chart to show open, approved and closed requests.
- **Add to a Dashboard:** Publish the report to a dashboard so the IT manager can monitor turnaround time at a glance.

Milestone 5: Testing and Deployment

In this milestone, the flow is verified end to end on the developer instance and then packaged as an Update Set so it can be deployed to another instance.

Activity 5.1: Test the Solution

- **Impersonate the Requester:** Use the user menu to impersonate the demo requester and open the Service Portal at /sp.
- **Approved Path:** Order a Standard Laptop, impersonate the manager and approve the request. Confirm the catalog task is created, the request state changes and the requester is notified.
- **Rejected Path:** Order another laptop, reject the approval as the manager and confirm the request is closed and the requester is notified.
- **Check Flow Executions:** Open Flow Designer > Executions to see each step of the run and confirm there are no errors.



The screenshot shows the ServiceNow Catalog Items list. The table displays various items, with 'Standard Laptop' highlighted. The table columns include Name, Short description, Active, Roles, Catalogs, Category, Price, Type, and Updated.

Name	Short description	Active	Roles	Catalogs	Category	Price	Type	Updated
Standard Laptop	Lenovo - Carbon x1	true		Service Catalog	Hardware	\$1,100.00	Item	2022-11-20 20:46:33
StarTech Mini Display Port to DVI Adapter	PC / Mac Compatible	false		Service Catalog	Peripherals	\$17.23	Item	2022-11-20 20:46:33
StarTech Mini Display Port to VGA Adapter	PC / Mac Compatible	true		Service Catalog	Peripherals	\$24.50	Item	2022-11-20 20:46:33
StarTech USB Mini Hub	4 Port, USB 2.0	true		Service Catalog	Peripherals	\$11.55	Item	2022-11-20 20:46:33
StarTech USB to DVI Adapter	Multi Monitor Adapter	true		Service Catalog	Peripherals	\$49.99	Item	2022-11-20 20:46:33
STM MacBook Air Sleeve	STM Mac Sleeve - 13"	true		Service Catalog	Peripherals	\$24.26	Item	2022-11-20 20:46:33
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Temporary group membership	Request for temporary group membership	true		Service Catalog	Application and Account Access	\$0.00	Item	2022-05-01 23:59:53

Test Case	Steps	Expected Result
Approved request	Order laptop, manager approves	Catalog task created, state Work in Progress, approval email sent
Rejected request	Order laptop, manager rejects	Request closed incomplete, rejection email sent
Missing mandatory field	Submit without delivery location	Form blocks submission

Activity 5.2: Export the Update Set

- **Set the Update Set:** Make sure all changes were captured in a single Update Set (for example IT Procurement Laptop Automation).
- **Mark as Complete:** Go to All > System Update Sets > Local Update Sets, open your Update Set and set its state to Complete.
- **Export to XML:** Use Export Update Set (XML) to download the file.
- **Import on Another Instance:** On the target instance, go to Retrieved Update Sets, import the XML, then Preview and Commit.

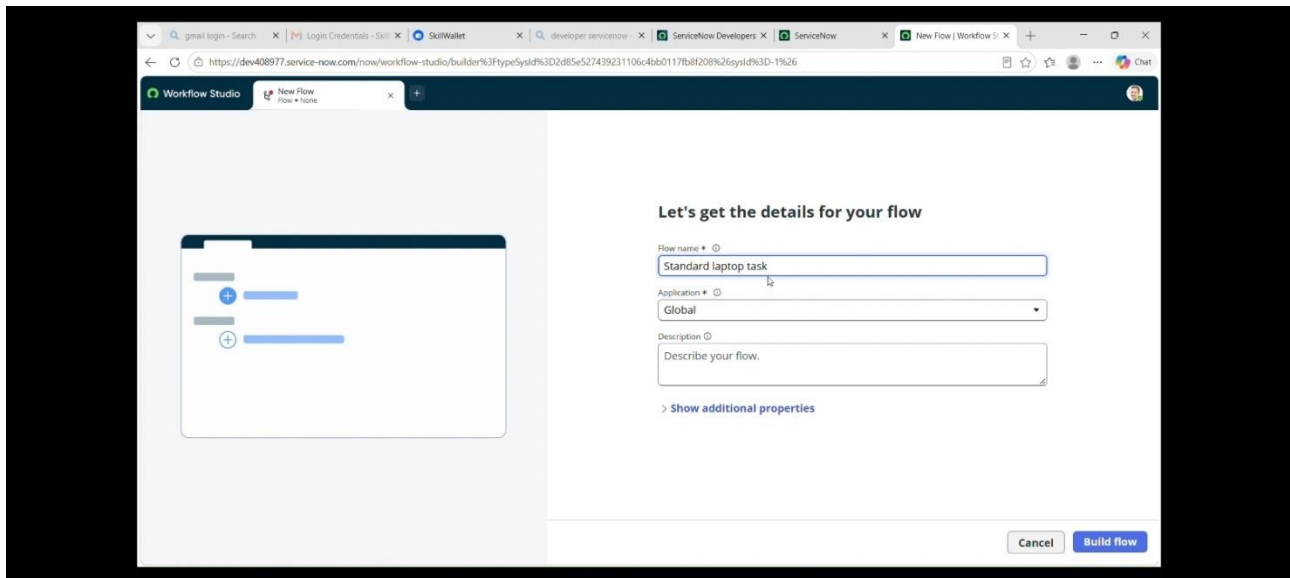
Important Notes:

- A developer instance hibernates after a period of inactivity. Wake it from the developer site before testing.
- Use demo users for testing and for the evaluator; never share the admin password.
- Test after every change to the flow, and re-activate the flow if it was edited.

Exploring the Solution's Pages:

Service Portal Catalog Page

Description: The Service Portal is the entry point for employees. The Standard Laptop item appears under the Hardware category and opens the request form in one click.



Workflow Studio | New Flow

Let's get the details for your flow

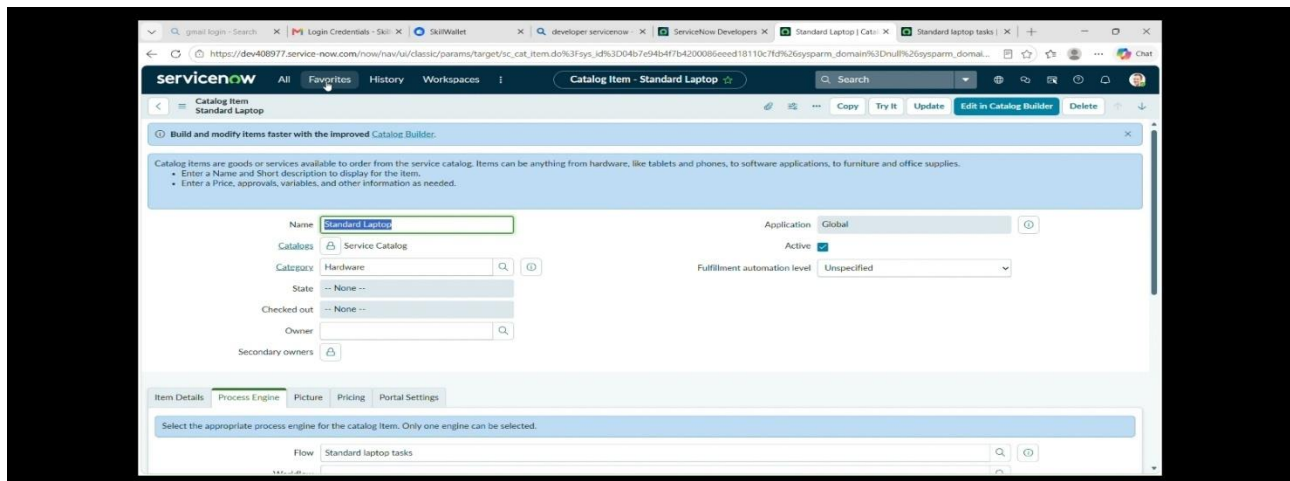
Flow name *

Application *

Description

[Show additional properties](#)

Request Form:



servicenow | All | Favorites | History | Workspaces | Catalog Item - Standard Laptop

Build and modify items faster with the Improved Catalog Builder.

Catalog Items are goods or services available to order from the service catalog. Items can be anything from hardware, like tablets and phones, to software applications, to furniture and office supplies.

- Enter a Name and Short description to display for the item.
- Enter a Price, approvals, variables, and other information as needed.

Name

Catalogs

Category

State

Checked out

Owner

Secondary owners

Application

Active ☒

Fulfillment automation level

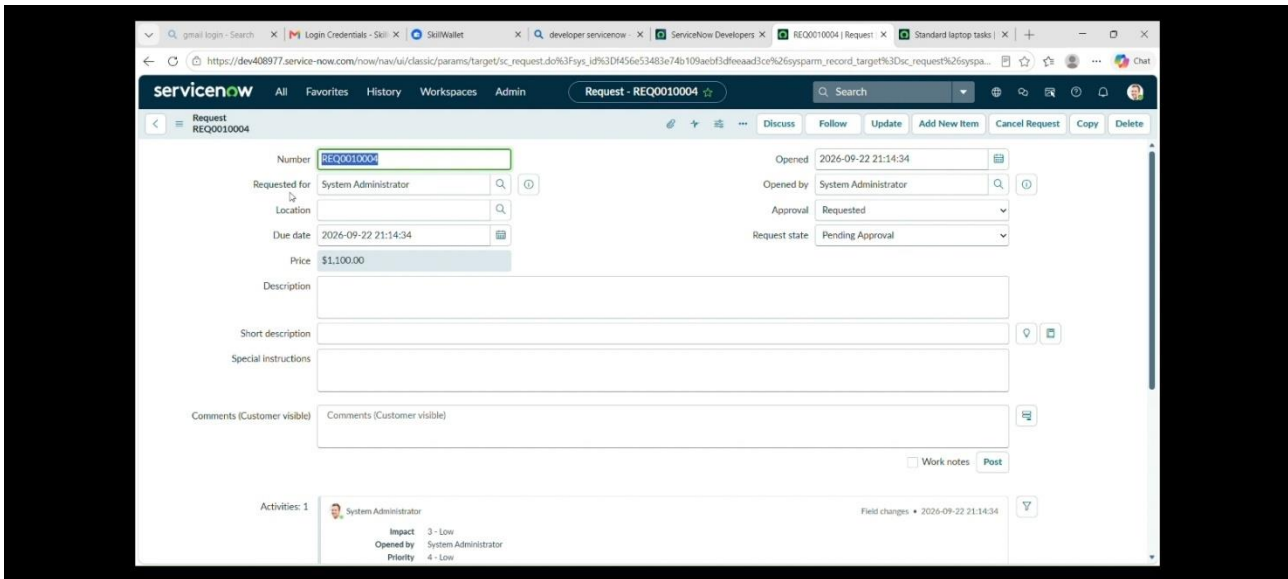
Item Details | Process Engine | Picture | Pricing | Portal Settings

Select the appropriate process engine for the catalog item. Only one engine can be selected.

Flow

Description: A simple form where the employee selects the laptop model, enters the business justification and delivery location, and clicks Order Now. Mandatory fields prevent incomplete requests.

Approval and Fulfillment Views:

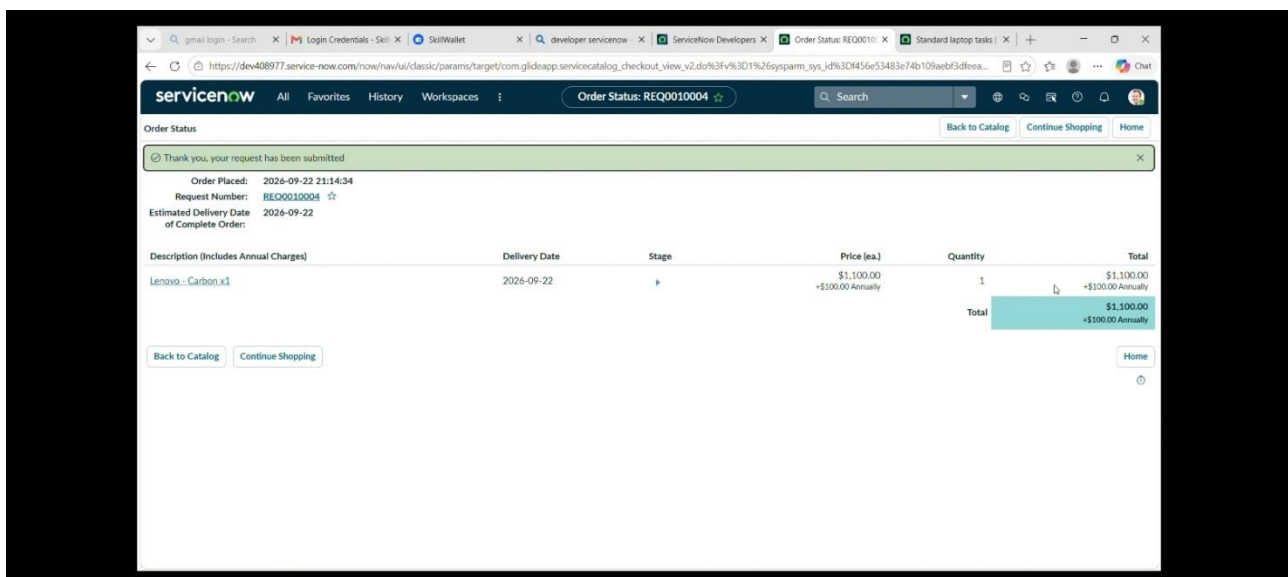


The screenshot shows the ServiceNow Request form for REQ0010004. The form is titled "Request - REQ0010004" and includes a search bar and navigation tabs (All, Favorites, History, Workspaces, Admin). The form fields are as follows:

- Number:** REQ0010004
- Requested for:** System Administrator
- Location:**
- Due date:** 2026-09-22 21:14:34
- Price:** \$1,100.00
- Description:**
- Short description:**
- Special instructions:**
- Comments (Customer visible):**
- Opened:** 2026-09-22 21:14:34
- Opened by:** System Administrator
- Approval:** Requested
- Request state:** Pending Approval
- Activities:** 1 (System Administrator)
- Field changes:** 2026-09-22 21:14:34

Description: The manager sees the pending approval with the request details and can approve or reject it. The decision moves the flow to the next step automatically.

Status Dashboard:



The screenshot shows the ServiceNow Order Status dashboard for REQ0010004. The dashboard includes a "Thank you, your request has been submitted" message and a table of order details.

Description (Includes Annual Charges)	Delivery Date	Stage	Price (ea.)	Quantity	Total
Lenovo - Carbon x1	2026-09-22		\$1,100.00 +\$100.00 Annually	1	\$1,100.00 +\$100.00 Annually
Total					\$1,100.00 +\$100.00 Annually

Description: The dashboard displays requests by state and shows how quickly laptop orders are completed, helping IT identify delays.

Conclusion:

This project shows how ServiceNow Service Catalog and Flow Designer can streamline IT procurement by automating standard laptop orders from request to fulfillment. A single catalog item, a manager approval, an automatic fulfillment task and timely email notifications replace manual emails and spreadsheets, cutting turnaround time and reducing errors. Role-based access and platform tables give a clear audit trail, and the report and dashboard give IT teams visibility of every order. Looking ahead, the solution can be extended with vendor or procurement system integration through IntegrationHub, automatic asset record creation on delivery, multi-level approvals based on cost, and support for more hardware such as monitors and phones. By reusing the same approval and task steps, the platform can grow from a laptop ordering process into a complete IT procurement service.